Lecture 7: Planetary Surfaces
Geology, Geomorphology, & Geophysics

Planetary Systems

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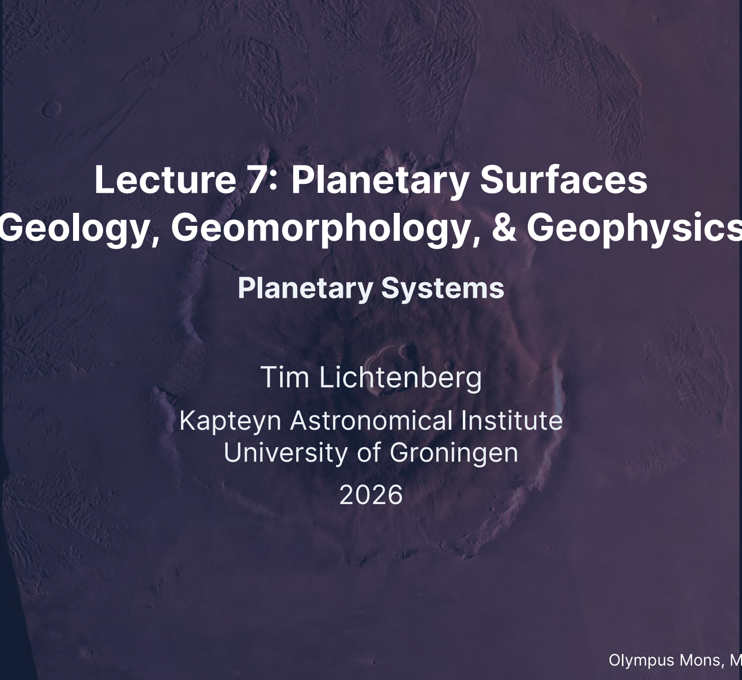


Olympus Mons, Mars. Credit: NASA/JPL/USGS

Learning Objectives

By the end of this lecture, you will be able to:

- Distinguish endogenic and exogenic surface processes
- Describe the three stages of impact crater formation
- Derive the crater scaling law $D \sim (E/\rho g)^{1/4}$ from dimensional analysis
- Use crater counting to determine relative and absolute surface ages
- Compare volcanic, tectonic, and erosional styles across the solar system
- Explain cryovolcanism on icy ocean worlds

Recap: Lectures 5 & 6

- Atmospheric structure: hydrostatic equilibrium, scale height, vertical layers
- Energy balance and greenhouse effect: $T_s > T_{\rm eff}$ from IR absorption
- Clausius-Clapeyron equation: cloud formation across the solar system
- Atmospheric dynamics: Hadley cells, Coriolis effect, jet streams
- Carbonate-silicate cycle: Earth's long-term climate thermostat
- Faint young Sun paradox; Venus runaway greenhouse; Mars climate puzzle

Today: What shapes the *surfaces* of planets and moons?

1. Surface Processes

Mars surface, Curiosity rover. Credit: NASA/JPL-Caltech/MSSS

Endogenic vs. Exogenic Processes

A planet's surface is its geological record: billions of years of competing processes that create, modify, and destroy landforms.

Endogenic (internal heat):

- Volcanism
- Tectonics (faulting, folding)
- More vigorous for larger, hotter bodies

Exogenic (external forces):

- Impact cratering
- Erosion (wind, water, ice)
- Space weathering, radiation

The balance between these processes determines what we see on a surface.

Dominant Surface Processes by Body

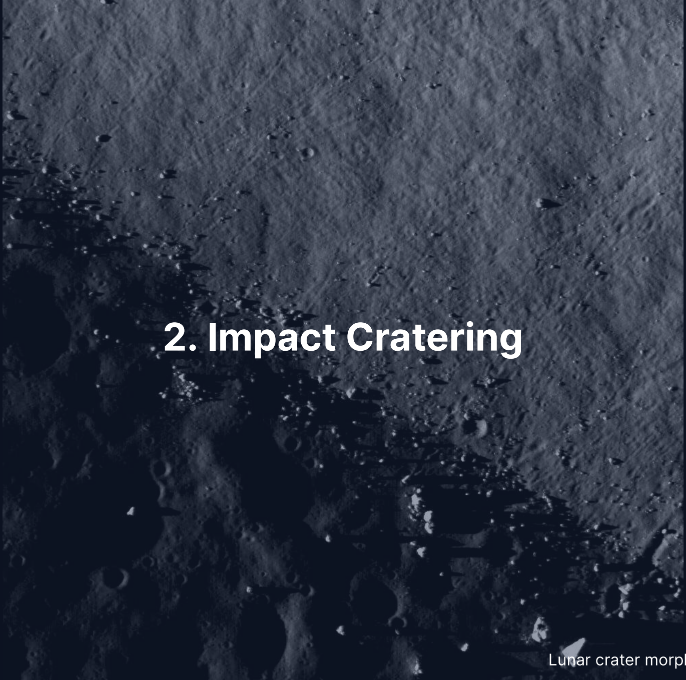
Body	Dominant process	Key evidence
Moon	Impact cratering	Saturated highland craters, >4 Gyr
Earth	Plate tectonics + erosion	Ocean floors <200 Myr; rapid weathering
Mars	Volcanism + impacts	Olympus Mons; cratered southern highlands
Venus	Volcanism	Volcanic plains cover ~80% of surface
Io	Tidal volcanism	~400 active volcanoes; age <1 Myr
Europa	Ice tectonics	Lineae, chaos terrain, very few craters

Internal heat → endogenic processes dominate.
Dead interior → surface preserves ancient craters.

Reading the Surface Record

Credit: NASA/JPL-Caltech/MSSS

- Aeolian dunes sculpted by wind in Gale Crater, Mars
- Wind erosion dominates Mars today
- Ancient volcanism and impacts shaped the deeper record
- On Earth, erosion destroys craters within ~10 Myr



2. Impact Cratering

Lunar crater morphology. Credit: NASA/GSFC/ASU

The Most Universal Geological Process

- **Every** solid body in the solar system bears impact craters
- On bodies without atmospheres or geology (Moon, Mercury), craters survive billions of years
- Impact velocities: 10–70 $\mathrm{km\,s^{-1}}$ (set by orbital mechanics)
- Enormous energy release in a fraction of a second

Three stages of crater formation:

1. Contact and compression
2. Excavation
3. Modification

Stage 1: Contact and Compression

- Projectile strikes at 10–70 km s^{-1}
- Shock wave propagates into both target and impactor
- Peak pressures ~100 GPa (comparable to Earth's core–mantle boundary)
- Material near impact point is **vaporised and melted**
- Kinetic energy → internal energy in < 1 s

The kinetic energy of an impactor:

$$E_k = \frac{1}{2}mv^2$$

Example: 1 km Asteroid Impact

Rocky asteroid: $\rho \approx 3000\ \mathrm{kg\,m^{-3}}$, diameter = 1 km

$$m = \frac{4}{3}\pi r^3 \rho \approx \frac{4}{3}\pi(500)^3 \times 3000 \approx 1.6 \times 10^{12}\ \mathrm{kg}$$

At $v = 20\ \mathrm{km\,s^{-1}}$:

$$E_k = \frac{1}{2} \times 1.6 \times 10^{12} \times (2 \times 10^4)^2 \approx 3 \times 10^{20}\ \mathrm{J}$$

~100× the energy of the largest nuclear weapon ever detonated, released in <1 s at a single point.

Stages 2 & 3: Excavation and Modification

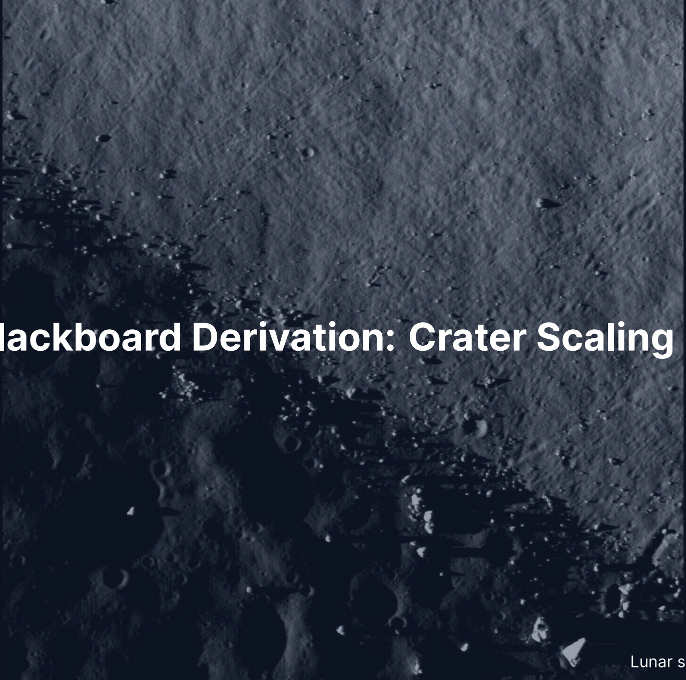
Excavation:

- Shock wave expands hemispherically outward
- Rarefaction wave accelerates material upward and outward
- **Transient cavity** forms (depth:diameter ≈ 1:3)
- Cone-shaped **ejecta blanket** deposited around the crater

Modification:

- Transient cavity is gravitationally unstable
- Small craters: walls slump inward → **simple crater** (bowl-shaped)
- Large craters: floor rebounds, walls collapse in terraces → **complex crater** with central peak

3. Blackboard Derivation: Crater Scaling Law



Lunar surface. Credit: NASA/GSFC/ASU

Setup: Crater Size from Dimensional Analysis

Goal: How does crater diameter D depend on impact energy E, target density ρ, and surface gravity g?

In the **gravity regime** (craters ≳100 m, where gravity limits size):

$$D = C\, E^{a}\, \rho^{b}\, g^{c}$$

Dimensions (M, L, T):

$$\begin{aligned} [D] &= L \\ [E] &= ML^{2}T^{-2} \\ [\rho] &= ML^{-3} \\ [g] &= LT^{-2} \end{aligned}$$

Solving for the Exponents

Require $[E^a \rho^b g^c] = L$:

Mass: $a + b = 0 \implies b = -a$

Time: $-2a - 2c = 0 \implies c = -a$

Length: $2a - 3b + c = 1$

Substituting $b = -a$ and $c = -a$:

$$2a + 3a - a = 1 \implies 4a = 1 \implies a = \frac{1}{4}$$

Therefore $a = 1/4$, $b = -1/4$, $c = -1/4$:

$$\boxed{D \sim \left(\frac{E}{\rho g}\right)^{1/4}}$$

Interpretation of the Scaling Law

- D scales as the **fourth root** of energy
- Doubling the energy increases D by only $2^{1/4} \approx 1.19$ (19%)
- This explains the relatively narrow range of crater sizes even though impactor energies span many orders of magnitude

Dimensional check:

$$\left[\frac{E}{\rho g}\right] = \frac{ML^2T^{-2}}{ML^{-3} \cdot LT^{-2}} = L^4 \quad \Rightarrow \quad [D] = L \ \square$$

The more complete **pi-scaling framework** (Holsapple & Schmidt, 1993) parameterises the transition between the gravity and strength regimes.

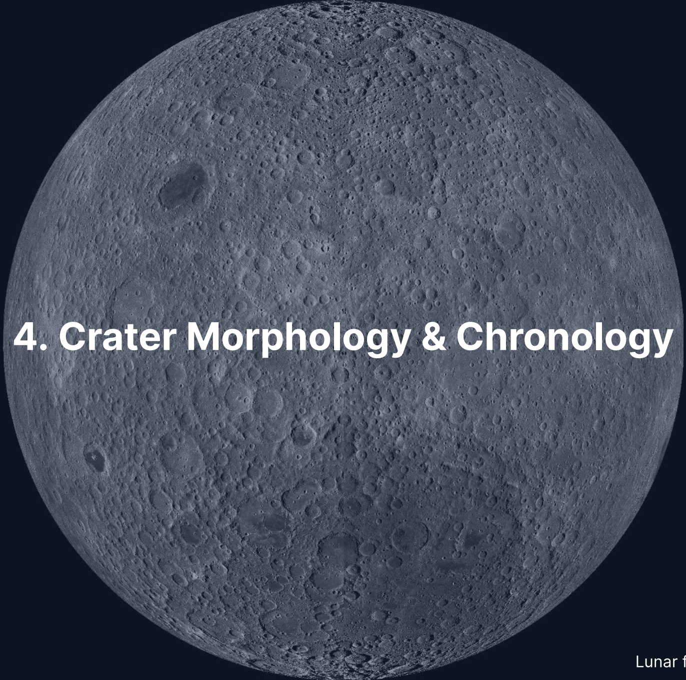
Worked Example: 1 km Asteroid on the Moon

Parameter	Symbol	Value
Impact energy	E	3×10^{20} J
Regolith density	ρ	$2500\ \mathrm{kg\,m^{-3}}$
Surface gravity	g	$1.62\ \mathrm{m\,s^{-2}}$

$$D \sim \left(\frac{3 \times 10^{20}}{2500 \times 1.62}\right)^{1/4} = (7.4 \times 10^{16})^{1/4} \approx 16.5\ \mathrm{km}$$

Consistent with observed lunar craters from ~1 km impactors.
Since $D \propto L^{3/4}$ at fixed v and ρ, the 85 km crater Tycho implies a ~9 km impactor (order of magnitude only).

4. Crater Morphology & Chronology



Lunar farside. Credit: NASA/GSFC/ASU

Crater Morphology: Three Classes

Credit: NASA/GSFC/ASU

1. **Simple** ($D < D_t$)
   Bowl-shaped; depth:diameter ≈ 1:5
   Moon: $D_t \approx 15$ km
2. **Complex** ($D_t < D \lesssim 300$ km)
   Central peak, terraced walls, flat floor
   Gravity overcomes wall strength
3. **Multi-ring basins** ($D \gtrsim 300$ km)
   Concentric ring structures
   Orientale (930 km), Caloris (1550 km), Hellas (2300 km)

Transition Diameter

The simple-to-complex transition scales with surface gravity:

$$D_t \approx D_{t,\mathrm{Moon}} \cdot \frac{g_{\mathrm{Moon}}}{g}$$

Body	g **(m s^{-2})**	D_t **(km)**
Moon	1.62	15
Mars	3.72	~6.5
Earth	9.81	~2.5
Mercury	3.70	~6.6

On Earth, virtually all craters larger than a few km are complex.

Crater Size-Frequency Distribution

The number of craters larger than a given diameter D follows an approximate power law:

$$N(>D) \propto D^{-b}, \quad b \approx 2\text{–}3$$

- Small craters are much more numerous than large ones
- Slope b depends on impactor size distribution and erosion
- A plot of $\log N$ vs. $\log D$ yields a straight line for a pristine surface
- Higher cumulative density $\Rightarrow$ older surface
- Deviations from the power law indicate resurfacing, saturation, or secondary craters

Source: Neukum et al. (2001), *Space Sci. Rev.*

Crater Counting: Principle

Key idea: Older surfaces have had more time to accumulate craters.

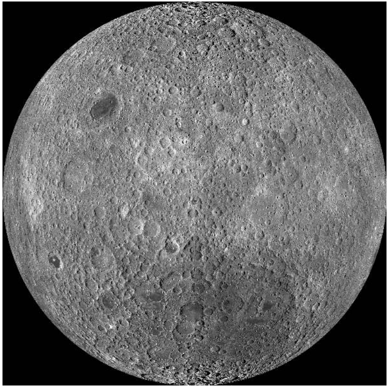
The **crater size-frequency distribution (SFD)** follows a power law:

$$N(> D) = a\,D^{-b}$$

- $N(> D)$: cumulative number of craters per unit area with diameter > D
- Production population: $b \approx 2$–3
- Plot N vs. D on log-log axes → straight line
- Higher line = older surface

Crater counting gives **relative ages**. Apollo/Luna samples provide **absolute calibration**.

The Lunar Chronology



Credit: NASA/GSFC/ASU

Apollo and Luna samples give radiometric ages for specific surfaces:

- **Highlands:** saturated, >4 Gyr
- **Maria:** 3.9–3.1 Gyr
- Calibrated SFD → absolute ages

Extrapolated to other bodies:

- Mars southern highlands: ~4 Gyr
- Mars northern lowlands: much younger
- Venus: uniform crater density → ~300–700 Myr

Key Results from Crater Chronology

- **Late Heavy Bombardment** (~3.9 Ga): spike in impact rate recorded on the Moon
- **Mars hemispheric dichotomy:**
  Southern highlands heavily cratered (~4 Gyr)
  Northern lowlands much smoother → volcanic resurfacing
- **Venus:** Remarkably uniform crater density
  Mean surface age ~300–700 Myr
  Implies a **global resurfacing event**
- **Europa:** Very few craters → surface age ~40–90 Myr
  Continuously resurfaced by ice tectonics

Break

Earthrise, Apollo 8 (1968). Credit: NASA

5. Volcanism

Olympus Mons, Mars. Credit: NASA/JPL/USGS

Effusive vs. Explosive Volcanism

The key variable is **magma viscosity**, controlled by $\mathrm{SiO_2}$ content:

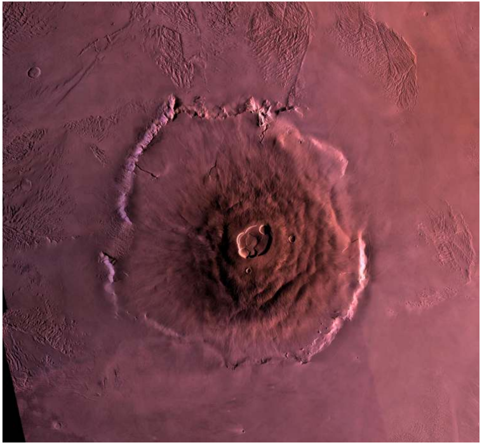
Low viscosity (basaltic):

- ~50% $\mathrm{SiO_2}$
- Flows easily
- Volatiles escape gradually
- **Effusive** eruptions
- Broad shield volcanoes, flood basalts
- Moon, Mars, Io

High viscosity (silicic):

- >65% $\mathrm{SiO_2}$
- Traps volatiles
- Pressure builds until failure
- **Explosive** eruptions
- Steep stratovolcanoes
- Primarily Earth

Olympus Mons: Largest Volcano in the Solar System



Credit: NASA/JPL/USGS

- Shield volcano on Mars
- Base diameter: ~600 km
- Height: ~21.9 km above surroundings
- 2.5× Everest above sea level
- Basal escarpment up to 6 km high

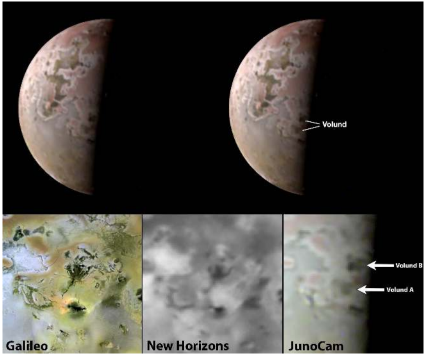
No plate tectonics on Mars → hotspot stays fixed → lava piles up for billions of years in one location.

Lunar Maria: Flood Basalts

- Vast basalt plains filling ancient impact basins on the nearside
- Erupted between 3.9 and 3.1 Ga (radiometric ages from Apollo samples)
- Cover ~16% of the lunar surface
- Visible from Earth as the dark patches forming the “face” of the Moon
- Record a period of residual internal heating, now extinct

On Earth, plate motion carries the crust over hotspots → chains of smaller volcanoes (e.g., Hawaiian Islands) rather than single massive edifices.

Venus & Io: Active Volcanic Worlds



Credit: NASA/JPL-Caltech

Venus:

- Lava plains cover ~80% of surface
- >1600 volcanic centres
- Stagnant lid; possible episodic overturns

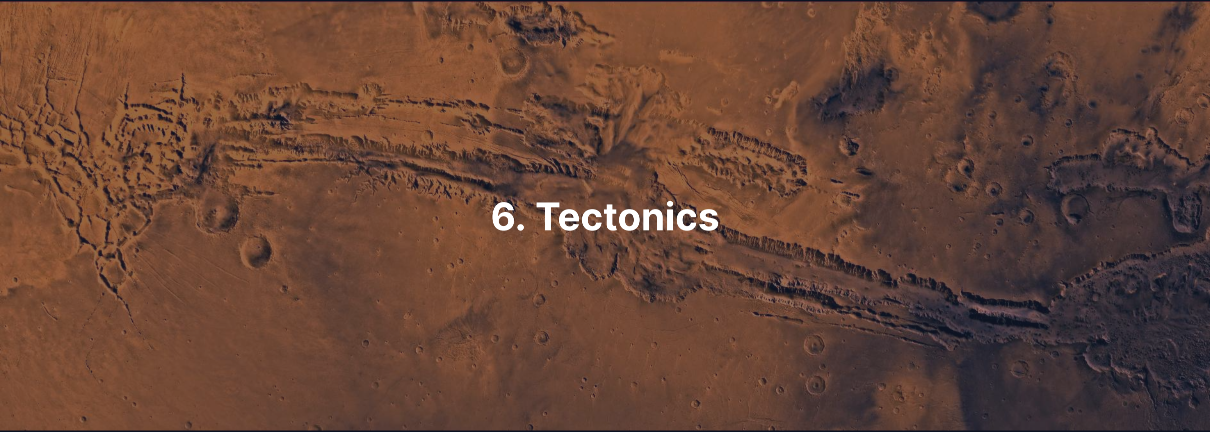
Io:

- Most volcanically active body
- ~400 active volcanic centres
- Powered by tidal heating (Laplace resonance)
- Surface age <1 Myr

Volcanism Across the Solar System

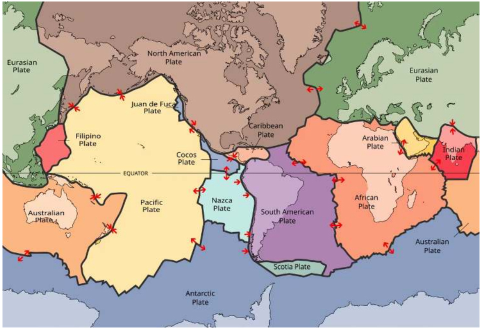
Body	Style	Driving mechanism	Examples
Earth	Effusive + explosive	Internal heat + recycling	Hawaii, mid-ocean ridges
Moon	Flood basalts (extinct)	Residual heat (3.9–3.1 Ga)	Maria
Mars	Shield volcanoes	Mantle plumes, no plates	Olympus Mons, Tharsis
Venus	Lava plains + shields	Internal heat, stagnant lid	Maat Mons
Io	Ultramafic flows	Tidal heating	Loki Patera, Pele

6. Tectonics



Valles Marineris, Mars. Credit: NASA/JPL/USGS

Plate Tectonics: Earth's Unique Regime



Credit: Wikimedia, CC BY-SA 3.0

Earth is the **only** body with active plate tectonics:

- ~15 rigid plates, moving at 1–10 cm yr^{-1}
- **Divergent:** mid-ocean ridges (new crust)
- **Convergent:** subduction zones (recycling)
- **Transform:** horizontal sliding

Enables the **carbonate-silicate cycle**: Carbon recycling → long-term climate regulation.

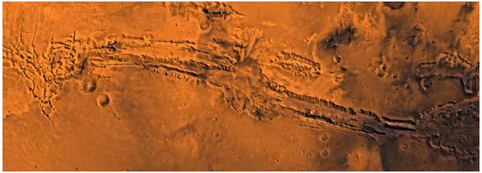
Stagnant-Lid Convection

- All other terrestrial bodies: **single rigid lithospheric lid**
- Mantle convects beneath, but lid does not participate
- Heat escapes by conduction through the lid and occasional volcanism
- Lid thickens over time as interior cools → volcanism eventually shuts down

Why is Earth unique?

- Viscosity contrast at lid base: ~ 10^{10}
- Breaking the lid requires water weakening + self-sustaining damage
- Likely a combination of size, water content, and thermal history

Mars: Tharsis and Valles Marineris



Credit: NASA/JPL/USGS

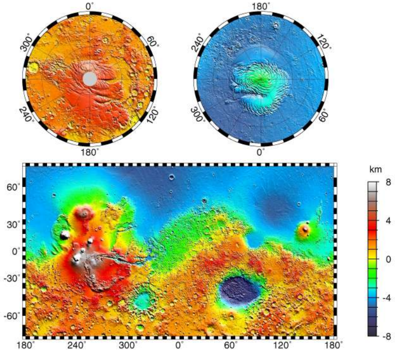
Tharsis bulge:

- Volcanic plateau ~5000 km across, ~10 km high
- Likely uplifted by a mantle plume

Valles Marineris:

- Largest canyon system: ~4000 km long
- Up to 7 km deep, 200 km wide
- Extensional rifting from Tharsis
- Lisbon to Moscow in scale

Mars Topography: The Hemispheric Dichotomy



Credit: NASA/JPL/GSFC (MOLA)

~6 km elevation difference between southern highlands and northern lowlands. Tharsis bulge with 4 giant volcanoes; Hellas basin (–8.2 km); Valles Marineris at equator.

Venus & Mercury: Contrasting Tectonic Styles

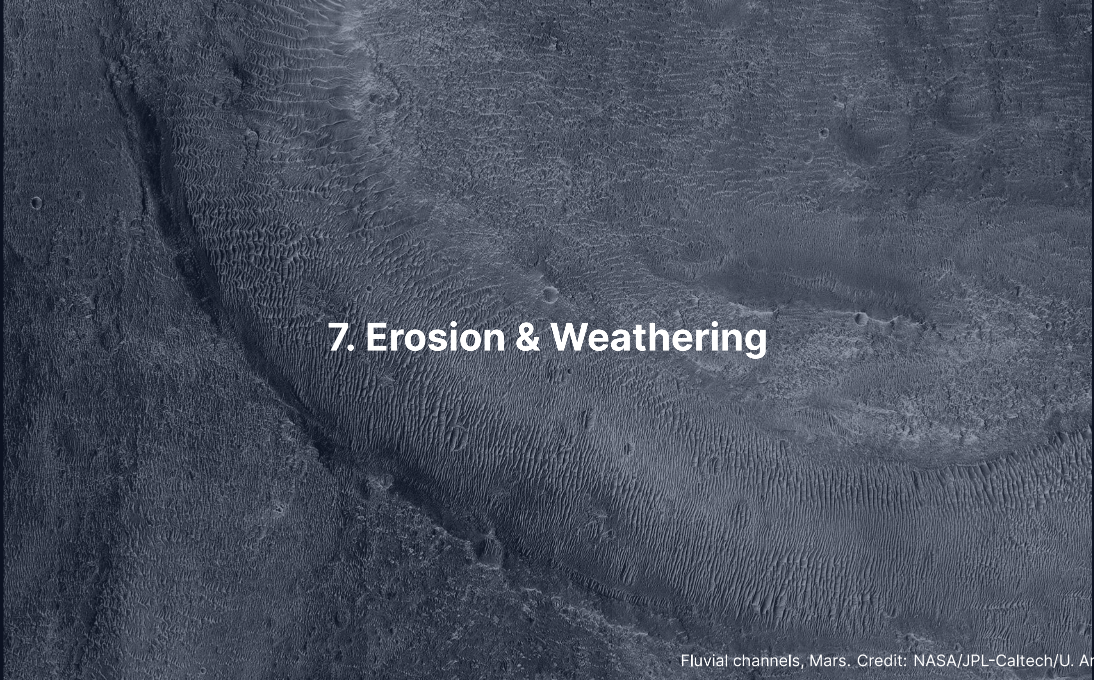
Venus, episodic resurfacing:

- Uniform crater density → mean age ~300–700 Myr
- Hypothesis: periodic catastrophic lid foundering
- Entire surface resurfaced volcanically in a short interval
- Between episodes: tectonically quiet

Mercury, global contraction:

- Large iron core cooled and solidified over time
- Planet's radius shrank by ~7 km
- Produced **lobate scarps**: thrust faults up to several hundred km long, 1–3 km high
- Mapped by Mariner 10 and MESSENGER

7. Erosion & Weathering



Fluvial channels, Mars. Credit: NASA/JPL-Caltech/U. Arizona

Aeolian (Wind) Processes

Wind erosion and deposition require a sufficiently dense atmosphere:

- **Mars:** Extensive dune fields in craters and polar regions
  Global dust storms loft particles to 30 km altitude
  Active dune migration observed by Curiosity
- **Titan:** Vast equatorial dune fields of organic particles (tholins)
  Longitudinal dunes up to 150 m tall, hundreds of km long
- **Venus:** Dense atmosphere but slow surface winds (~1 $m\,s^{-1}$)
  Even slow winds can mobilise fine particles in thick atmosphere

Fluvial (Water) Processes

Credit: NASA/JPL-Caltech/U. Arizona

Mars valley networks:

- Noachian-aged (>3.7 Ga)
- Dendritic drainage patterns resembling terrestrial rivers
- Evidence for sustained liquid water flow

Outflow channels:

- Hundreds of km long
- Catastrophic groundwater releases

Titan: Active methane rivers; Huygens imaged rounded ice pebbles in a dry riverbed (2005).

Glacial & Chemical Weathering

Glacial processes:

- Earth: ice sheets covered ~30% of land at Last Glacial Maximum (~20 ka)
- Mars: polar $\mathrm{CO_2}$ and $\mathrm{H_2O}$ ice caps; mid-latitude features resemble rock glaciers

Chemical weathering:

- Earth: silicate weathering by carbonic acid is the critical carbon sink
- Mars: orbital spectroscopy (CRISM, OMEGA) detected hydrated minerals: phyllosilicates, sulfates, carbonates
- Venus: high surface T (~737 K) drives surface–atmosphere reactions

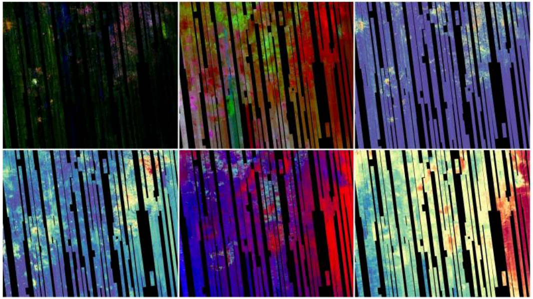
Remote Sensing of Surfaces: Overview

We can visit only a few sites. Most surface knowledge comes from orbital remote sensing.

Reflectance spectroscopy (UV–IR) identifies minerals via absorption bands
Radar imaging penetrates clouds and regolith, measures surface roughness
Laser altimetry maps topography at sub-metre vertical precision
Gravity field mapping constrains subsurface density anomalies

Each technique probes a different length scale and physical property; together they build a three-dimensional picture of the surface and shallow interior.

Reflectance Spectroscopy: Mineral Mapping



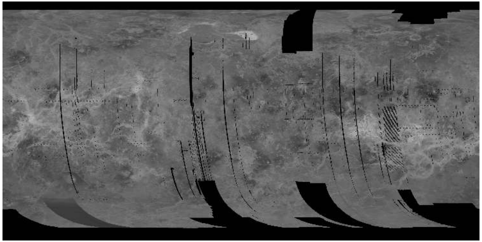
Credit: NASA/JPL-Caltech/JHU APL

CRISM on Mars Reconnaissance Orbiter:

- Maps minerals from near-infrared absorption features
- Fe/Mg phyllosilicates (clays)
- Olivine, carbonates
- Formed by aqueous alteration of basalt

Strongest mineralogical evidence for past liquid water on Mars.

Radar Imaging: Seeing Through Clouds



Credit: NASA/JPL Magellan Cycle 1

Active microwave imaging is insensitive to clouds and darkness.

Magellan at Venus (1990–94):

- SAR mapped 98% of the surface at ~100 m resolution
- Revealed coronae, tesserae, lava flows, young cratered surface

Cassini at Titan (2004–17):

- Ku-band radar pierced the organic haze
- Mapped hydrocarbon lakes and seas

Radar is the only way to image surfaces shrouded by thick atmospheres.

Laser Altimetry and Gravity Mapping

Laser altimetry (MOLA, LOLA, BELA):

- Time-of-flight of laser pulses from orbit
- Vertical precision ~1 m over global baselines
- MOLA produced the canonical Mars topographic map (used throughout this lecture)

Gravity field mapping (GRAIL at the Moon, GRACE/GOCE at Earth):

- Spacecraft-to-spacecraft tracking measures tiny accelerations
- Gravity anomalies reveal buried mass: dense cores, thin crust, basins, mantle plumes
- Combined with topography, constrains crustal thickness and compensation state

Altimetry and gravity together tell us what the interior **looks like** from above.

Regolith & Space Weathering

Credit: NASA/Apollo 11

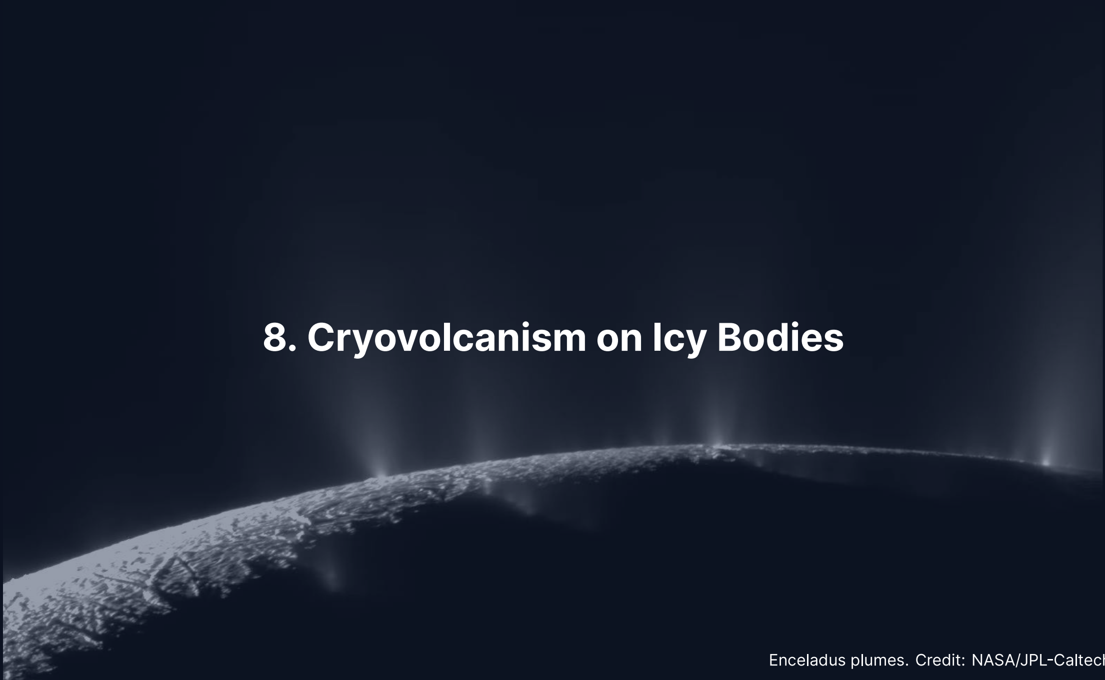
Regolith:

- Loose, fragmented surface layer on airless bodies
- Produced by billions of years of impact gardening
- Depth: 2–4 m maria, 6–8 m highlands

Space weathering:

- Solar wind ion implantation
- Micrometeorite melting → nanophase iron
- Surfaces become **darker and redder**
- Fresh craters appear bright (e.g., Tycho)

8. Cryovolcanism on Icy Bodies



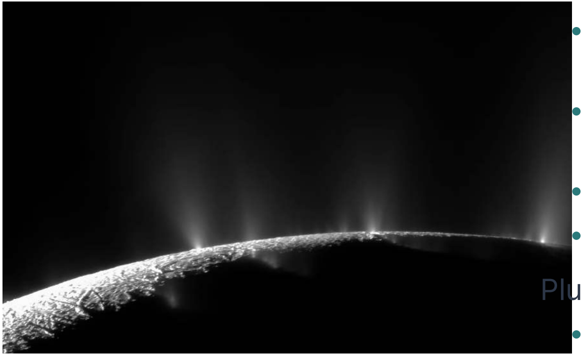
Enceladus plumes. Credit: NASA/JPL-Caltech/SSI

Cryovolcanism: Ice World Volcanism

In the outer solar system, “volcanic” eruptions involve volatiles rather than silicate melts:

- Erupted materials: liquid water, ammonia–water mixtures, methane, nitrogen
- Energy source: **tidal heating** maintains subsurface oceans beneath icy shells
- “Magma” = low-viscosity volatile liquids
- Analogous to silicate volcanism, but at temperatures of 100–270 K

Enceladus: Tiger Stripes and Geysers



Credit: NASA/JPL-Caltech/SSI (Cassini)

- Radius only 252 km
- Water ice jets from 4 parallel “tiger stripe” fractures
- Sourced from **global subsurface ocean**
- Ocean in contact with rocky core
- Thermal emission: ~15 GW

Plume composition:

- H_2, silica nanoparticles, complex organics
- Consistent with hydrothermal vents

Europa: A Hidden Ocean

- Jupiter's moon, $R \approx 1561$ km
- Global ocean ~100 km deep; ice shell ~10–30 km thick
- Maintained by tidal heating (Laplace resonance with Io and Ganymede)

Surface features:

- **Lineae:** cracks filled by upwelling ocean water or warm ice
- **Chaos terrain:** broken, rotated, refrozen ice blocks
- **Very few craters:** surface age ~40–90 Myr

Europa Clipper (launched 2024): dozens of close flybys to characterise the ice shell, ocean, and habitability.

Triton & the Ocean Worlds

Triton (Neptune's largest moon):

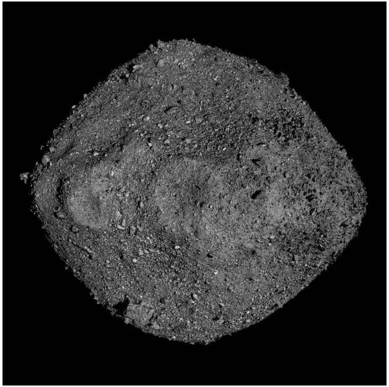
- Nitrogen geysers observed by Voyager 2 (1989)
- Plumes of N_2 gas and dark dust, rising ~8 km
- Very young surface; retrograde orbit (captured KBO)
- Possible subsurface ocean

Enceladus, Europa, and Triton demonstrate that liquid water can exist far beyond the classical habitable zone, maintained by tidal heating rather than stellar radiation.

9. Recent Advances

Mars surface. Credit: NASA/JPL-Caltech/MSSS

OSIRIS-REx: Rubble-Pile Asteroids



Credit: NASA/GSFC/University of Arizona

NASA's OSIRIS-REx returned ~70 g of asteroid **Bennu** in September 2023.

- ~500 m rubble-pile asteroid
- Near-Earth orbit, carbonaceous (B-type), water-rich
- Hydrated clays, carbonates, phosphates
- Amino acids and nucleobases
- Extremely porous: a "loose pile of rocks"
- Insight into early solar system aqueous alteration

Source: Lauretta et al. (2019, 2024)

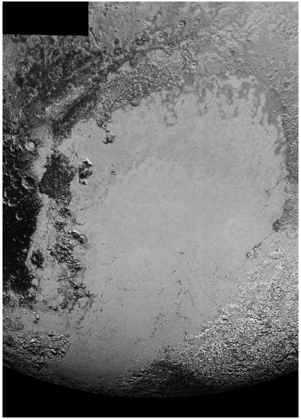
Hayabusa2: Sampling Ryugu

JAXA's Hayabusa2 delivered 5.4 g of asteroid Ryugu to Earth in December 2020.

- C-type carbonaceous asteroid, 900 m diameter
- First sample of a water-bearing asteroid returned to Earth
- Contains pristine organic molecules: amino acids, sugars, uracil
- Evidence for aqueous alteration on the parent body
- Noble gas and rare-earth abundances constrain its formation location
- Complements OSIRIS-REx by sampling a different asteroid

Source: Watanabe et al. (2019); Yada et al. (2022)

Pluto: A Surprisingly Active World



Credit: NASA/JHUAPL/SwRI (New Horizons 2015)

New Horizons (2015) revealed Pluto's geological diversity.

- **Sputnik Planitia:** 1000 km nitrogen ice glacier
- Cellular convective patterns: active N_2 ice convection today
- Surface age $\lesssim$ 10 Myr
- Mountains of water ice up to 3 km tall
- Possible subsurface liquid water ocean

Source: Stern et al. (2015); Moore et al. (2016)

Perseverance Rover at Jezero Crater

- Operating in Jezero crater since February 2021
- Crater floor: igneous rock (olivine-bearing cumulates)
- Subsequently altered by liquid water
- >20 sample tubes cached for Mars Sample Return
- Joint NASA/ESA mission: first laboratory analysis of Martian rocks
- Will address past habitability and possible biosignatures

Jezero was chosen because it is an ancient lake crater with a preserved delta deposit, maximising the chance of finding evidence for past life.

DART: Planetary Defence Experiment

- **Double Asteroid Redirection Test** (September 2022)
- Kinetic impactor deliberately crashed into moonlet **Dimorphos**
- Changed orbital period around Didymos by **33 minutes**
- First successful planetary defence experiment
- Confirmed kinetic impact as a viable deflection strategy
- Impact ejecta revealed properties of rubble-pile asteroid surfaces

DART demonstrated that we can deflect a potentially hazardous asteroid using existing technology.

Source: Daly et al. (2023), *Nature*; Thomas et al. (2023)

Ongoing Exploration: Io and Venus

Io monitoring:

- Juno extended mission: close Io flybys in 2023–2024
- Discovery of previously unknown eruption sites
- Constraints on spatial distribution of tidal heat flow
- Ground-based adaptive optics complement spacecraft data

Upcoming Venus missions:

- VERITAS and EnVision (planned for early 2030s)
- First high-resolution surface radar data since Magellan (1990–1994)
- Will test whether Venus has recent or ongoing volcanic activity

Summary

1. Surfaces record the competition between **endogenic** (volcanism, tectonics) and **exogenic** (impacts, erosion) processes
2. Impact cratering is universal; three stages: contact, excavation, modification
3. Crater scaling law: $D \sim (E/\rho g)^{1/4}$ (fourth-root dependence on energy)
4. Crater counting + Apollo calibration → absolute surface ages
5. Volcanism style depends on magma viscosity; no plate tectonics → giant volcanoes
6. Earth is unique in having plate tectonics; other bodies have stagnant lids
7. Fluvial features on Mars: strong evidence for past liquid water
8. Cryovolcanism on Enceladus, Europa, Triton: liquid water beyond the habitable zone

Questions?

Next (after mid-term exam): **Lecture 8. Planetary Interiors: Structure, Composition, & Dynamics**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`